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MILITARY SPECIFICATION

PRINTER, PHOTOGRAPHIC, AUTOMATIC FOCUS, 4 BY 5-INCH, TYPE EN-52( )

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification states the requirements for an auto-focus projection printer for making enlargements from roll and sheet films, 35mm to 5 inches in width.

1.2 Classification. The projection printer shall be of one type, designated EN-52( ). (See 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-1	Standard for Laboratory Atmospheric Condition for Testing

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-150	Photographic Lenses
MIL-STD-152	General Environmental Requirements, Testing Procedures, and Sampling for Aerial Photographic Reconnaissance Equipment
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ENGINEERING SPECIFICATIONS AND STANDARDS DEPARTMENT (CODE 93) NAVAL AIR ENGINEERING CENTER, LAKEHURST, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6740

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MIL-STD-462	Electromagnetic Interference Test Requirements and Test Methods
MS26565	Spool, Photographic Film, Aerial

SPECIFICATIONS

FEDERAL

NN-P-530	Plywood, Flat Panel
PPP-B-591	Box, Fiberboard, Wood Cleated
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-675	Coating of Glass Optical Elements (Antireflection)
MIL-C-3767/12	Connector, Plug, Electrical (Power Three-Wire, Polarized, Spring-Loaded, Pivoted Grounding Blade) Type UP131M
MIL-C-3767/13	Connector, Plug, Electrical (Power Three-Wire, Fixed Grounding Blade), Type UP221M
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

(Copies of standards, specifications, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H-28	Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents Government Printing Office, Washington, D. C. 20402).

AMERICAN NATIONAL STANDARDS INSTITUTE

PH1.14	35mm Film Magazines and Film for Still Picture Cameras
PH1.21	Roll Film, Backing Paper, and Spools
PH3.31	Photographic Enlargers, Methods for Testing

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018).

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3. REQUIREMENTS

3.1 Preproduction. This paragraph establishes the requirement for preproduction testing. (See 6.2.1)

3.2 Design. The printer shall be capable of producing 1.5X to 17.5X photographic enlargements from roll or sheet film negatives onto sensitized photographic materials by means of an automatic focusing vertical optical system with interchangeable condensers and lenses. The printer shall accommodate film widths from 35mm to 5 inches with nominal format sizes of 18 by 24mm up to and including 4-1/2 by 4-7/8 inches.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be of good commercial quality entirely suitable for the purpose for which they are used. Materials shall be free from all defects and imperfections that might affect the serviceability or appearance of the finished product.

3.3.1 Screw threads. Screw threads employed in the fabrication of the printer shall be of standard commercial sizes conforming to the requirements of the National Bureau of Standards Handbook H-28.

3.3.2 Corrosion resistance. The printer shall be made of inherently corrosion-resistant materials or materials suitably treated to resist corrosion. The treatment used shall not interfere with the operation of the printer. Protective coatings shall not crack, scale or powder when operated under the environmental conditions specified herein.

3.3.3 Fungus-inert materials. Materials which are not nutrients for fungus shall be used to the greatest extent practicable. Where materials that are nutrients must be used, such materials shall be treated with a fungicidal agent. The fungus proof materials and treatments used shall be selected and applied in accordance with MIL-STD-152. Bellows made of vinyl materials, and lenses shall be acceptable without fungicidal treatment.

3.4 Interchangeability. Similar units, parts and accessories of each printer shall be functionally interchangeable, without modification, with corresponding units, parts and accessories of other printers furnished on the same contract.

3.5 Rigidity and alignment. Each part of the printer shall be of sufficient strength and rigidity to accomplish its function and to maintain smooth operation with proper fit and alignment during the life of the printer. The design and construction of the printer shall minimize and quickly dampen vibratory movements of the projector assembly.

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3.6 Maintenance. The printer shall be constructed to allow maintenance without the use of special tools. Wherever practicable, parts shall permit removal and replacement without interference from, damage to, or removal of other parts or wiring.

3.7 Dimensions. Overall dimensions of the printer and base shall not exceed 20 inches wide by 36 inches long. The maximum operating height shall not exceed 60 inches above the base.

3.7.1 Tolerances. Unless otherwise specified all tolerances on dimensions shall be plus or minus 1/8-inch.

3.8 Environmental conditions.

3.8.1 Operating. The printer shall operate satisfactorily in any ambient temperature from plus 55°F to plus 95°F and withstand exposure to relative humidity up to 95 percent.

3.8.2 Storage. The printer shall withstand, without damage, storage in any ambient temperature from minus 80°F to plus 160°F.

3.8.3 Shock and vibration. The printer shall withstand, without damage, the shocks and vibration encountered during shipment by air, rail, or truck. When installed for use, the printer shall withstand, without damage, the Environmental Vibrations (Type I) Tests of MIL-STD-167 except that the frequency range shall be 5 to 25 cycles per second.

3.8.4 Electromagnetic interference. The design and construction of the printer shall conform to the general requirements specified in MIL-STD-461 for Class II equipment.

3.9 Temperature rise. The temperature rise at the negative plane shall not exceed 20 degrees F above ambient.

3.10 Uniformity of illumination. The illumination over the entire printing plane, with the printer set for maximum enlargement, shall be uniform within the limits specified herein. The intensity of illumination at any corner of the projected format shall be not less than 60 percent of the intensity at the center. The intensity of illumination at any corner shall not vary more than ten percent from the average of the intensities at the four corners. The entire projected format area shall be free from visually perceptible color bands or local differences in illumination.

3.11 Major components. The printer shall include the following major components:

Baseboard
Supporting column
Projection assembly
Easel, if specified (See 6.2)

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3.12 Baseboard. The baseboard shall be made of hardwood plywood, Group A, in accordance with Specification NN-P-530, with dimensions of 1-1/8 by 18 by 36 inches ± 1/8-inch. The surfaces of the baseboard shall be smooth, flat and free from warping that could affect the stability or alignment of the printer.

3.13 Supporting column. The supporting column shall consist essentially of rigid metal structural members of suitable configuration to serve as tracks on which the carriage can be raised or lowered. The column shall be inclined approximately fifteen degrees from the optical axis and the tracks shall be accurately positioned in a manner to maintain alignment of the optical axis of the projection assembly, perpendicular to the baseboard within plus or minus 15 minutes of arc, with the carriage at any elevation on the column. Means shall be provided on the column for holding three interchangeable auto-focusing cam tracks in proper alignment with the focusing mechanism. The column shall be firmly attached to the baseboard. The column may be removed without the use of special tools. Means shall be provided on the column for attaching a brace to fasten the printer top to a wall or the ceiling.

3.14 Projection assembly. The projection assembly shall consist of a movable carriage containing an automatic focusing optical system capable of projecting an enlarged image of a transilluminated negative onto the printing plane. The projection assembly shall include a counterbalanced carriage, lamphouse, condensing lenses, focusing mechanism, negative holders and objective lenses.

3.14.1 Counterbalance. The projection assembly's stability on the column shall be maintained by means of spring loaded counterbalances. The balances shall be adjusted to check any marked tendency of the carriage to move in either the up or down direction with any of the negative holders or lenses in place without applying the brake. The counterbalance springs shall be easily accessible for adjustment or replacement without the use of special tools.

3.14.2 Carriage. The projection assembly shall be mounted on a rigidly constructed movable carriage to facilitate adjustment of the image magnification. The carriage shall be mounted on the column by means of smooth running rollers. The rollers shall be arranged to distribute the weight of the carriage and maintain the accurate alignment of the optical system at all positions on the column.

3.14.2.1 Elevating mechanism. The carriage shall be raised and lowered on the supporting column by means of a manually operated rack and pinion elevating mechanism controlled by a conveniently located hand wheel.

3.14.2.2 Brake. The carriage shall be provided with a manually operated friction brake to hold the projection assembly securely at any selected elevation on the column. The brake shall not mar or damage the metal of the supporting column.

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3.14.2.3 Manual focusing mechanism. The carriage shall be fitted with a manual focusing mechanism to provide for fine focusing of the projected image. There shall be no backlash in the fine focusing mechanism. All moving parts shall be constructed and aligned to minimize wear.

3.14.2.4 Automatic focusing mechanism. The projection assembly shall be fitted with an automatic focusing mechanism consisting of a system of levers actuated by a rolling cam follower that engages the cam track corresponding to the objective lens being used. The focusing mechanism shall maintain the projected test target image at the printing plane in sharp focus throughout the entire travel of the projection assembly, when tested as specified in section 4 of his specification. Moving parts shall be constructed in a manner to minimize wear and backlash that will affect the accuracy of focus. The cam follower shall be quickly and easily adjustable, without the use of tools, to permit accurate engagement with any of the cam tracks on the column.

3.14.2.4.1 Cams. Three interchangeable auto-focusing cam tracks shall be furnished with the printer. The cams shall be precision matched to the focusing characteristics of the objective lenses furnished with the printer and each cam shall be suitably marked with the focal length and serial number of the lens with which it shall be used. The cams shall be of the floating, self adjusting type whereby inaccuracies of the column are not transferred to the focusing movement. The cams shall be attachable and removable from the column without the use of special tools.

3.14.3 Lamp house. The printer shall have a removable lamp house containing the light source, variable condenser, and fixed condensers, respectively. The lamp house shall be removable without the use of special tools to permit replacement with other types of illumination, such as, electronic flash, fluorescent lamps, variable contrast light source, and color head if specified in the procurement document (See 6.2). Means shall be provided for lifting and locking the lamp house in a raised position to facilitate changing negative holders and moving the film. The lamp house shall be lighttight to the extent that stray light from the lamp house shall not fog sensitized material.

3.14.3.1 Upper housing. The upper section of the lamp house shall be a lighttight enclosure for the lamp and equipped with cooling fans and light baffled ventilating ports. The lamp shall be a 150 watt photographic enlarging lamp, General Electric Number PH/212 or equal.

3.14.3.2 Central housing. The center section of the lamp housing shall be a lighttight enclosure with facilities for holding heat absorbing glass, color filters and an auxiliary condensing lens. The housing shall have a lighttight access door to facilitate changing filters and adjusting the condensing lens. The means to be provided for holding the adjustable condensing lens element shall ensure proper centering and proper spacing, with respect to the fixed condensers and the light

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source, to obtain optimum performance with each of the objective lenses specified herein. A clearly legible diagram shall be permanently attached to the inside of the access door to show the proper position of the variable condenser for each of the objective lenses furnished with the printer. The center section shall contain the means for attaching the upper section and condenser assemblies to the lamp house and means for attaching the lamp house to the carriage.

3.14.3.3 Lower housing. The lower section of the lamp housing shall contain the condensing lens system consisting of three plano-convex lenses in accordance with type XI of MIL-STD-150. The lenses shall be in the form of a two-element fixed component and a single element variable component. The fixed condensing lenses shall be a minimum of seven inches in diameter and the variable condensing lens element shall be a minimum of 4-11/16 inches in diameter. The focal length, spacing of elements, and covering power of the condensing lens system shall be matched to the characteristics of the respective objective lenses with which its use is intended. The lenses of the fixed component shall be mounted in a tight metal cell suitable for attachment to the center section of the lamp house. The lens element of the variable component shall be suitable mounted to fit the center section of the lamp house. The mount shall engage a series of precalibrated positions in the housing in a manner to maintain proper spacing and optical alignment of the condensing system for each of the respective objective lenses.

3.14.4 Negative holders. When specified in the procurement document (see 6.2) the printer shall be furnished with one or more of the following types and sizes of negative holders or with negative holders as specified in the procurement document. The holders shall be interchangeable without the use of tools and shall provide the means to ensure accurate alignment of the aperture with the optical system.

3.14.4.1 4 by 5-inch holder. The holder shall accommodate nominal 4 by 5-inch film or film pack negatives. The film shall be held flat between glass plates.

3.14.4.2 5-inch aerial roll film holder. The holder shall accommodate 250 foot lengths of 5-inch aerial roll film negatives wound on spools conforming to MS26565-9. The film shall be held flat between glass plates. The printing aperture shall conform to figure 1. The holder shall be equipped with rewinds to facilitate advancing and rewinding the film.

3.14.4.3 2-1/4 by 3-1/4-inch holder. The holder shall accommodate 120 size 2-1/4 by 3-1/4-inch roll film negatives in accordance with ANSI PH1.21. The film shall be held flat between glass plates. Suitable troughs shall be provided to support uncut lengths of roll film.

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3.14.4.4 70 millimeter holder. The holder shall accommodate 100 foot lengths of 70 millimeter aerial roll film negatives wound on spools conforming to MS26565-3. The film shall be held between glass plates. The printing aperture shall conform to figure 2. The holder shall be equipped with rewinds to facilitate advancing and rewinding the film.

3.14.4.5 2-1/4 by 2-3/4-inch glassless holder. The holder shall accommodate 15 foot lengths of unspooled 70 millimeter roll film negatives between metal plates with an aperture of 2-1/4 by 2-3/4-inches. Suitable troughs shall be provided to support the uncut lengths of film during printing.

3.14.4.6 2-1/4 by 2-1/4-inch glassless holder. The holder shall accommodate 15 foot lengths of unspooled 70 millimeter roll film negatives, between metal plates with a 2-1/4 by 2-1/4-inch aperture. Suitable troughs shall be provided to support the uncut lengths of film during printing.

3.14.4.7 35 millimeter glassless holder. The holder shall accommodate 36 exposure lengths of 35mm roll film conforming to ANSI PH1.14. The film shall be held between metal plates with a suitable printing aperture for use with a nominal 24mm by 36mm (double frame) format. Suitable troughs shall be provided to support the uncut lengths of film during printing.

3.14.4.8 Half frame 35 millimeter glassless holder. The 35 millimeter half frame holder shall be identical with the double frame holder specified herein except that the printing aperture shall be suitable for use with negatives of nominal 18mm by 24mm format.

3.14.5 Objective lenses. Three interchangeable objective lenses shall be furnished with the printer. The lenses shall be type VI in accordance with MIL-STD-150. The focal length, relative aperture, and format covered, shall be as specified in Table I. The equivalent focal length of each lens shall be within plus or minus four percent of the nominal focal length specified in Table I. The minimum resolution for each lens supplied with the printer shall be as specified in Table I.

Table I

Focal Length	Relative Aperture	Enlargement Min.	Enlargement Max.	Test Format	Test Magnification	Minimum Resolution (Lines per mm)
50mm	f/4.5	5x	17.5x	24 x 36mm	8 diameters	40
105mm	f/5.6	1.5x	7.5x	2-1/4 x 2-1/4 in.	4 diameters	32
162mm	f/5.6	1.5x	4x	4-1/2 x 4-1/2 in.	2 diameters	26

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Each lens shall have a between-the-lens iris diaphragm calibrated in the f/stop system and an illuminated diaphragm setting ring fitted with click stops at each marked aperture. Each lens shall be mounted on a lens board or cone in a manner to ensure proper alignment with the optical system of the printer. The lens board shall be easily interchangeable without the use of tools.

3.14.5.1 Lens coating. All glass-air surfaces of the lenses shall have reflection reducing coatings in accordance with MIL-C-675.

3.14.5.2 Lens cap. Front and rear lens caps shall be furnished with each lens.

3.15 Film scratch. The printer shall not scratch or otherwise damage either the emulsion, base or edges of the film negatives in any manner during operation.

3.16 Easel. When specified by the procuring activity (See 6.2), an easel of specified size shall be furnished with the printer. The printing plane of the easel shall be 1-inch above the surface of the printer base-board.

3.17 Electrical. The printer shall operate from a nominal 115 VAC, plus or minus 5 volts, 50/60 cycle, single phase power supply.

3.17.1 Power cord. The printer shall be equipped with a ten-foot length of three conductor rubber-covered power cord, terminating in a three wire, grounding type connector conforming to MIL-C-3767/12 or MIL-C-3767/13. One wire of the cord shall be connected between the grounding blade of the plug and the metal frame of the lamp house.

3.18 Accessory case. A suitable accessory case shall be provided to hold the objective lenses that are not in use on the printer. The case shall have fitted compartments, suitably cushioned to provide protection for the accessories during shipment and storage.

3.19 Technical literature. A technical manual giving details for installation and operation shall be furnished with each printer. When specified in the procurement document (See 6.2), a complete maintenance and repair manual shall be furnished.

3.20 Identification marking. Identification marking of the printer shall be in accordance with MIL-STD-130.

3.20.1 Serial number. The contractor shall assign a serial number to each printer. This number shall not be duplicated for any similar piece of equipment.

3.21 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with the best commercial practice for the fabrication of precision projection printers.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Classification of tests. The printer shall be subjected to Preproduction Tests and Quality Conformance Tests.

4.3 Test conditions.

4.3.1 Temperature and humidity. Unless otherwise specified herein, the printer shall be tested under conditions of temperature and humidity as specified in FED-STD-No. 1.

4.3.2 Voltage. The printer shall operate on 115 VAC, plus or minus 5 volts, 50/60 cycle, single phase.

4.3.3 Lamps. Lamps used for the test shall be 150 watt General Electrical PH/212 Photo Enlarger lamps, or equal.

4.3.4 Test materials. The test materials shall be as specified in Method 14 of MIL-STD-150.

4.3.5 Test plate. The targets used for the resolving power tests shall consist of a series of standard high contrast elements with clear lines on a dark background, decreasing in size at the sixth root of two. Arrangement of the targets on the test plate shall conform to the requirements of Method 14 of MIL-STD-150 for a "Projected Resolving Power Test Plate".

4.4 Preproduction tests.

4.4.1 Preproduction samples. The preproduction samples shall be representative of the production model in design, construction and workmanship. The number of printers to be selected for preproduction testing, and the testing agency shall be designated by the procuring activity. (See 6.2.1)

4.4.2 Preproduction inspection. The preproduction inspection shall consist of all of the tests specified herein. The tests are listed in Tables II and III.

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Table II
Preproduction tests

Test	Test Paragraph	Requirement Paragraph
Lens Tests	4.6.7	3.14.5
Lens Coating	4.6.7.1	3.14.5.1
Magnification Ratio	4.6.8	3.14.5
Film Scratch	4.6.9	3.15
Temperature Rise at Negative Plane	4.6.10	3.9
Rigidity	4.6.11	3.5
Environmental Tests:		
Humidity	4.6.12	3.8.1
Temperature	4.6.13	3.8.2
Shock and Vibration	4.6.14	3.8.3
Electromagnetic Interference	4.6.15	3.8.4
Fungus	4.6.16	3.3.3
Packaging and Packing	4.6.17	Section 5

*The preproduction tests shall also include the quality conformance tests listed in Table III.

4.5 Quality conformance tests.

4.5.1 Quality conformance tests. The tests listed in Table III shall be performed on each sample printer selected according to the applicable provisions of MIL-STD-105. Sample sizes shall be selected at general inspection level II with an acceptable quality level of 1.5. The lot size shall consist of all printers offered for delivery at one time on any one contract or order.

4.5.2 Classification of defects. Failure to pass any of the quality conformance tests shall be considered a major defect and cause for rejection.

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Table III
Quality conformance tests

Test	Test Paragraph	Requirement Paragraph
Physical Examination	4.6.1	Covers all requirement paragraphs for which no specific tests are listed
Lighttightness	4.6.2	3.14.3
Uniformity of Illumination	4.6.3	3.10
Resolving Power	4.6.4	3.14.5
Automatic Focusing	4.6.5	3.14.2.4
Vertical Alignment	4.6.6	3.13

4.6 Test procedures.

4.6.1 Physical examination. Examine the printer visually and with gages to determine conformance with the requirements of this specification with respect to design, size, materials, construction, fit of components, wiring, finish, marking, accessories, and workmanship. Determine that the baseboard is smooth and flat. Verify that the projection assembly can be properly counterbalanced with each specified combination of lens, condenser, and carrier assembly furnished with the printer. Determine that the elevating mechanism moves the carriage smoothly throughout the entire range of adjustment. Verify that the brake holds the carriage securely in place at any elevation. Inspect the ventilating port baffles and condenser access door for evidence of light leaks. Check the dimensions and alignment of the negative holders and determine that they operate properly with the respective sheet film, roll film, and film spool for which each holder is specified.

4.6.2 Lighttightness. The printer shall be tested in accordance with the applicable procedures of ANSI PH3.31 to determine conformance with the requirements as specified.

4.6.3 Uniformity of illumination. This test shall be performed utilizing the largest format negative carrier and corresponding objective lens. With the printer set at maximum magnification and the lens at maximum aperture and no film in the negative carrier, divide the

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projected image area of the negative carrier at the printing plane into a rectangular grid pattern of at least 25 squares. Measure the incident illumination at the printing plane at the center of each square with a suitable photometer having a sensing element smaller than the area of a single grid element. Determine that the illumination at any corner is not less than 60 percent of the intensity at the center, and the intensity at any corner does not vary more than ten percent from the average of the intensities at all four corners. Examine the projected illumination visually for evidence of perceptible color bands or local variations in illumination. Repeat the test with each of the lenses furnished with the printer.

4.6.4 Resolving power. The resolving power of the printer shall be determined in accordance with method 14 of MIL-STD-150. Perform the tests at the relative apertures, negative formats, and test magnifications specified in Table I. Determine that the minimum resolving power for the 50mm, 105mm, and 162mm lenses, at any point within the test format at the printing plane is not less than 40, 32, and 26 lines per millimeter, respectively.

4.6.5 Automatic focusing. With the lens set at maximum aperture, visually examine the projected image of the test plate at the printing plane by means of a suitable optical instrument, such as a focusing magnifier with a clear glass reticle. Observe the test target image at the printing plane while raising and lowering the projection assembly throughout the entire range of travel and verify that the test target image does not depart from sharp focus throughout the entire range of travel of the projection assembly.

4.6.6 Vertical alignment. By means of a suitably calibrated spirit level, verify that the negative plane, lens mounting surface, and baseboard of the printer are parallel with each other over the entire range of adjustment of the projection assembly. Insert a negative containing mutually perpendicular parallel lines, or a rectangular grid pattern in the negative carrier and measure the projected image on the easel for lack of parallelism or rectilinear distortion attributable to misalignment of the optical system.

4.6.7 Lens tests. Verify the equivalent focal length and relative aperture of each of the objective lenses is as specified in Table I when tested in accordance with the applicable procedures of MIL-STD-150.

4.6.7.1 Lens coating. Inspect the coated surfaces of the lenses in accordance with the applicable non-destructive test procedures of MIL-C-675 for conformance with the requirements specified therein. Where destructive testing is required, the tests shall be conducted on sample lens elements coated for test purposes by identical methods as those used in production of the lenses offered for acceptance.

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4.6.8 Magnification ratio. With an appropriately calibrated rectangular grid pattern in the negative plane of the printer, accurately measure the image at the printing plane and verify that the minimum and maximum magnification ratio obtainable with each lens conforms to the requirements as specified in Table I.

4.6.9 Film scratch. Place a fresh sheet or roll of film which has been directly removed from the manufacturer's package in the appropriate negative holder. Operate the printer in the normal manner and remove the film for evaluation. Roll film used for this test shall be advanced in accordance with the printer manufacturer's instructions. All film subjected to this test shall be examined for scratches and damage to the emulsion and base side of the film, and the edges of the film.

4.6.10 Temperature rise at the negative plane. Place a test negative with a uniform density of not less than 2.0 in the negative plane. Attach a suitable contact-type thermocouple to the center of the emulsion side of the negative. Turn the light source on and off alternately at ten second intervals for a period of 30 minutes. Record the temperature shown by the thermocouple immediately at the completion of the test period. The temperature rise shall not exceed 20 degrees F above the ambient temperature.

4.6.11 Rigidity. With the printer properly focused to produce a 17X enlargement of the resolution target plate, and an unexposed paper in place on the easel, slightly tilt the printer by inserting a block of suitable thickness to raise the front edge of the baseboard 3/4 inch. Remove the block suddenly, allowing the baseboard to fall flat onto the supporting surface in order to simulate vibration of the projection assembly resulting from normal use. Wait three seconds for vibrations to dampen and immediately expose the print. Compare the processed image with a print exposed prior to the initiation of the procedure to simulate vibration. Verify that no degradation of the projected test target image is visible.

4.6.12 Humidity. Conduct the humidity test in accordance with procedure III of MIL-E-5272. Upon return to nominal room conditions, wipe the printer dry, inspect for evidence of rust, corrosion, or other damage. Operate the printer and check all components for evidence of malfunction resulting from the humidity test.

4.6.13 Temperature tests. Store the printer for 24 hours at a temperature of plus 160 F, plus or minus 5 degrees F, then return to 95 F and conduct the operational tests. During and after each test period, examine the printer for evidence of malfunction or damage resulting from the temperature tests.

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4.6.14 Vibration. With the baseboard of the printer firmly attached to the vibration machine, conduct the tests for type I Environmental Vibration in accordance with the applicable requirements of MIL-STD-167, except that the frequency range shall extend only from 5 to 25 cycles per second. A head brace support shall be used during the test if the printer column has provisions for attaching a bulk-head brace. The printer shall not be operated during the test. At the conclusion of the tests, examine the printer for damage and determine that all components function properly.

4.6.15 Electromagnetic interference. The printer shall be tested in accordance with the applicable test methods of MIL-STD-462 specified for class II equipment.

4.6.16 Fungus test. With specimens of the materials used in construction of the printer which are nutrients to fungus, conduct the test in accordance with procedure I of MIL-E-5272, except that the test period shall be 14 days. At the end of the test period, examine the samples for growth of any fungus that might impair the proper operation of the printer.

4.6.17 Examination of packaging and packing. Examine the packaging and packing of the printer in accordance with the Quality Assurance Provisions of the specifications referenced in Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. The lenses and condensers shall be individually wrapped with an acid free and noncorrosive material, cushioned and placed in the accessory case. The complete printer, including the accessory case, shall be packaged in accordance with Method IC-2 of MIL-P-116, except that the contact preservative shall be omitted (see 6.2).

5.1.2 Level C. The lenses and condensers shall be individually wrapped with an acid free and noncorrosive wrapping, cushioned and placed in the accessory case. The complete printer, including accessory case, shall then be packaged in accordance with the manufacturer's commercial practice, and in a manner to afford protection against corrosion, deterioration and physical damage during direct shipment from the supplier to the user (see 6.2).

5.2 Packing.

5.2.1 Level A. Each printer, preserved and packaged level A as specified herein, shall be packed for shipment in a snug fitting class 2, Type 2, 2-1/2, or 3, overseas wood box for type 3 load in accordance

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with PPP-B-621; an overseas type, Style A or B cleated plywood box of type III, Class 1 plywood in accordance with PPP-B-601; or an overseas type, Style A or B wood cleated fiberboard box in accordance with PPP-B-591 (See 6.2).

5.2.2 Level B. Each printer, preserved and packaged, as specified in 5.1.1, shall be packed in domestic-type exterior shipping containers conforming to PPP-B-601 (style B), or PPP-B-621 (class 1, style 2). Containers shall be closed and strapped in accordance with the appendix of the applicable container specification. The gross weight of the container shall be limited to approximately 300 lbs (See 6.2).

5.2.3 Level C. Each printer, preserved and packaged level C as specified herein shall be packed for shipment in an exterior shipping container to insure carrier acceptance and safe delivery at the destination (See 6.2).

5.3 Marking. The interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The printer covered by this specification is intended for use in photographic darkrooms for producing enlarged prints from single photographic negatives in sheet and roll form. Negative sizes accommodated range from 35mm to 5 inches in width.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Number of printers required

(c) Additional lamp houses, if required (See 3.14.3)

(d) Negative holders required (See 3.14.5)

(e) Printing easel required (See 3.16)

(f) Complete maintenance and repair manual required (See 3.19)

(g) Levels of packaging and packing (See Section 5)

6.2.1 Preproduction sample. Contracts, purchase orders, and invitations for bids should specify the number of printers to be selected for preproduction testing, preproduction test conditions, the testing agency to which the sample printers should be submitted, instructions concerning submittal of test reports, and that the time of delivery of each sample and its test data be concurrent.

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6.3 Type designator. The parenthesis in the type designator will be replaced by a suffix letter and a production issue number. As soon as possible after award of the contract, the contractor should apply to the contracting officer for this information.

Custodians:
Army - EL
Navy - AS
Air Force - 99

Reviewer Activities
Army - EL
Navy -
Air Force - 99, 85

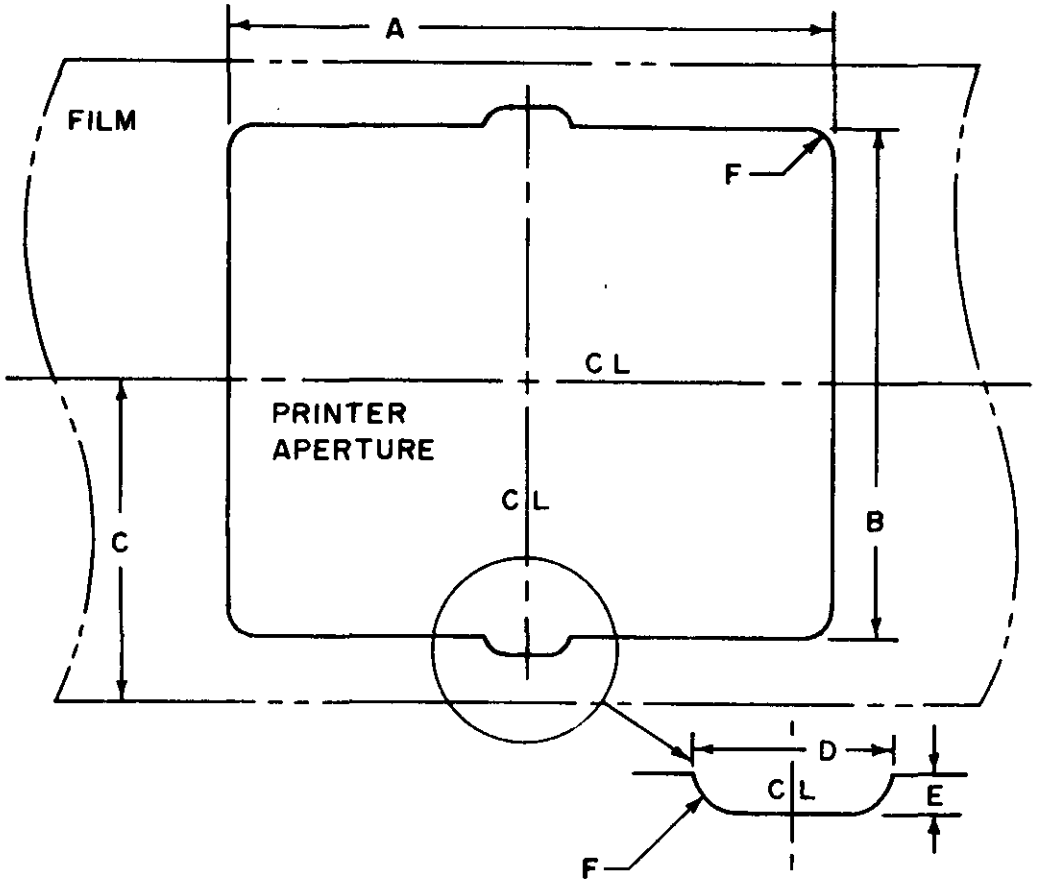
User Activities:
Army - EL
Navy - AS
Air Force - 99

Preparing Activity
Navy - AS

Project Number
6740-0194

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APERTURE FOR 5 INCH FILM



FORMAT DIMENSIONS

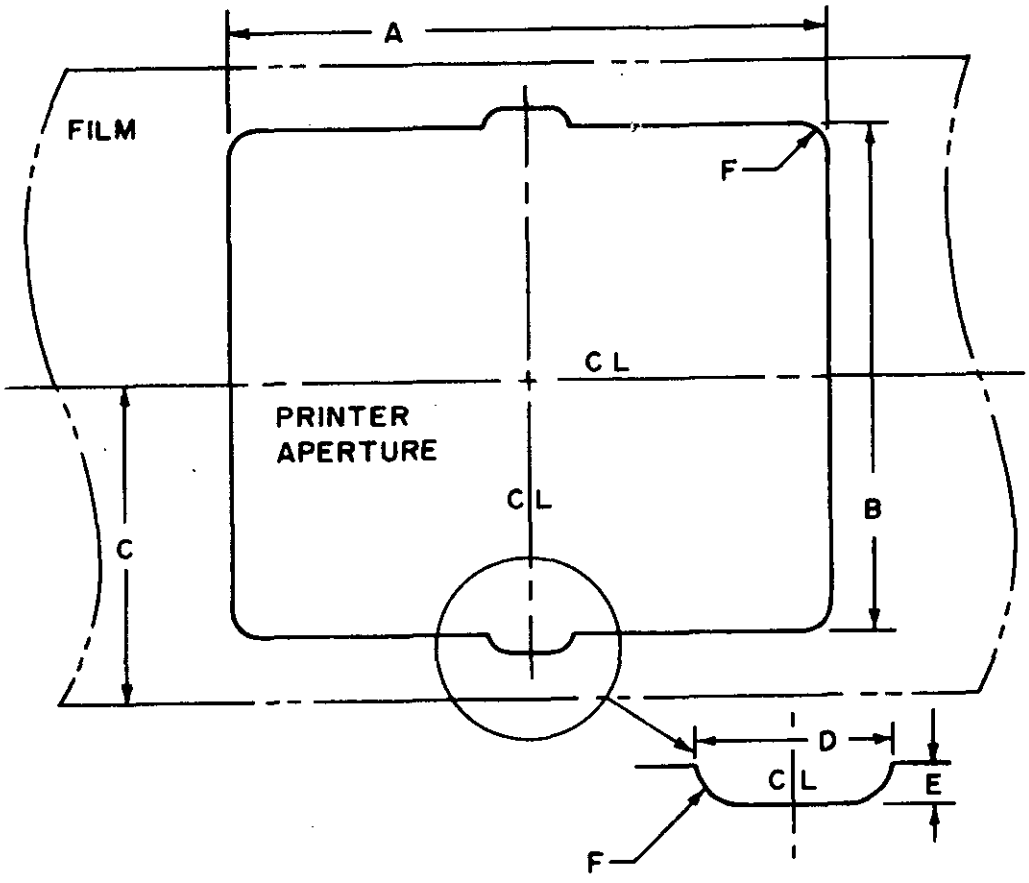
DIMENSIONS	INCHES			MILLIMETERS		
A	4.875	+0.000	−0.004	123.83	+0.00	−0.10
B	4.520	+0.004	−0.000	114.30	+0.00	−0.10
C	2.480			62.99		
D	0.750			19.05		
E	0.126			3.20		
F	0.126 Rad			3.20 Rad		

TOLERANCES: Unless otherwise specified ± .005 on dimensions shown in inches; ± .125mm on dimensions shown in millimeters.

FIGURE 1

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APERTURE FOR 70mm FILM



FORMAT DIMENSIONS

DIMENSIONS	INCHES			MILLIMETERS		
A	2.618	+0.000	−0.004	66.49	+0.00	−0.10
B	2.250	+0.000	−0.004	57.15	+0.00	−0.10
C	1.377			34.98		
D	0.375			9.53		
E	0.063			1.60		
F	0.063 Rad			1.60 Rad		

TOLERANCES: Unless otherwise specified ±.005 on dimensions shown in inches; ±.125mm on dimensions shown in millimeters.

FIGURE 2

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ENGINEERING SPECIFICATIONS AND STANDARDS DEPARTMENT
NAVL AIR ENGINEERING CENTER
LAKEHURST, NJ 08733

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Engineering Specifications and Standards Department(Code 93)
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Lakehurst, NJ 08733

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-P-7041D PRINTER, PHOTOGRAPHIC, AOTOMATIC FOCUS, 4 BY 5-INCH, TYPE EN-52( )

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

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